动态随机一般均衡模型：理论与应用
第 6 章：工资黏性

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A 动态系统

A.1 无指数化调整的情形

$$\beta\mathbb{E}_t\frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}}\frac{R_t}{\Pi_{t+1}}=1$$
$$\hat{\dot{W}}_t^{1+\epsilon_w\varphi}=\frac{1}{\mu_w}\frac{X_{1t}}{X_{2t}}$$
$$X_{1t}=\hat{W}_t^{\epsilon_w(1+\varphi)}N_t^{1+\varphi}+\beta\theta_w\mathbb{E}_t\Pi_{t+1}^{\epsilon_w(1+\varphi)}X_{1t+1}$$
$$X_{2t}=C_t^{-\sigma}\hat{W}_t^{\epsilon_w}N_t+\beta\theta_w\mathbb{E}_t\Pi_{t+1}^{\epsilon_w-1}X_{2t+1}$$
$$\hat{W}_t^{1-\epsilon_w}=\theta_w\frac{\hat{W}_{t-1}^{1-\epsilon_w}}{\Pi_t^{1-\epsilon_w}}+(1-\theta_w)\hat{\dot{W}}_t^{1-\epsilon_w}$$
$$\hat{W}_t/A_t=\mu$$
$$Y_t=A_tN_t$$
$$Y_t=C_t$$
$$\frac{R_t}{R}=\Pi_t^{\phi_\pi}$$
$$N_t^s=N_tV_t^w$$
$$V_t^w=(1-\theta_w)\big(\frac{\hat{\dot{W}}_t}{\hat{W}_t}\big)^{-\epsilon_w}+\theta_w\big(\frac{\hat{W}_{t-1}}{\hat{W}_t\Pi_t}\big)^{-\epsilon_w}V_{t-1}^w$$

A.2 存在指数化调整的情形

$$\beta\mathbb{E}_t\frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}}\frac{R_t}{\Pi_{t+1}}=1$$
$$\hat{\dot{W}}_t^{1+\epsilon_w\varphi}=\frac{1}{\mu_w}\frac{X_{1t}}{X_{2t}}$$
$$X_{1t}=\hat{W}_t^{\epsilon_w(1+\varphi)}N_t^{1+\varphi}+\beta\theta_w\Pi_t^{-\gamma_w\epsilon_w(1+\varphi)}\mathbb{E}_t\Pi_{t+1}^{\epsilon_w(1+\varphi)}X_{1t+1}$$
$$X_{2t}=C_t^{-\sigma}\hat{W}_t^{\epsilon_w}N_t+\beta\theta_w\Pi_t^{\gamma_w(1-\epsilon_w)}\mathbb{E}_t\Pi_{t+1}^{\epsilon_w-1}X_{2t+1}$$
$$\hat{W}_t=\Big[\theta_w\big(\Pi_{t-1}^{\gamma_w}\hat{W}_{t-1}/\Pi_t\big)^{1-\epsilon_w}+(1-\theta_w)\hat{\dot{W}}_t^{1-\epsilon_w}\Big]^{\frac{1}{1-\epsilon_w}}$$
$$\hat{W}_t/A_t=\mu$$
$$Y_t=A_tN_t$$
$$Y_t=C_t$$

$$\frac{R_t}{R} = \Pi_t^{\phi_\pi}$$
$$N_t^s = N_t V_t^w$$
$$V_t^w = (1-\theta_w)\left(\frac{\hat{W}_t}{\hat{W}_t}\right)^{-\epsilon_w} + \theta_w \left(\frac{\Pi_{t-1}^{\gamma_w}\hat{W}_{t-1}}{\hat{W}_t \Pi_t}\right)^{-\epsilon_w} V_{t-1}^w$$

B 求解模型稳态

B.1 稳态系统

设稳态时 $\Pi = 1$，那么

$$R = \hat{R} = \frac{1}{\beta}$$
$$N^\varphi C^\sigma = \mu_w \hat{W}$$
$$N = \frac{Y}{A}$$
$$\hat{MC} = \mu$$
$$\hat{MC} = \hat{W}/A$$
$$Y = C$$

B.2 模型稳态均衡

$$R = \hat{R} = \frac{1}{\beta}$$
$$\hat{W}_t = \mu A$$
$$Y = \left(\mu\mu_w\right)^{\frac{1}{\sigma+\varphi}} A^{\frac{1+\varphi}{\sigma+\varphi}}$$
$$C = \left(\mu\mu_w\right)^{\frac{1}{\sigma+\varphi}} A^{\frac{1+\varphi}{\sigma+\varphi}}$$
$$N = \left(\mu\mu_w\right)^{\frac{1}{\sigma+\varphi}} A^{\frac{1-\sigma}{\sigma+\varphi}}$$

C 对数线性化

C.1 无指数化调整的情形

$$c_t = \mathbb{E}_t c_{t+1} - \frac{1}{\sigma}(r_t - \mathbb{E}_t \pi_{t+1})$$
$$\pi_t^w = \hat{w}_t - \hat{w}_{t-1} + \pi_t$$
$$r_t = \phi_\pi \pi_t$$
$$\pi_t^w = \beta \mathbb{E}_t \pi_{t+1}^w + \frac{\kappa_w}{1+\epsilon_w\varphi}\left(\sigma c_t + \varphi n_t - \hat{w}_t\right)$$
$$y_t = a_t + n_t$$
$$\hat{w}_t = a_t$$
$$y_t = c_t$$

C.2 有指数化调整的情形

$$
\begin{aligned}
c_t &= \mathbb{E}_t c_{t+1} - \frac{1}{\sigma}(r_t - \mathbb{E}_t \pi_{t+1}) \\
\pi_t^w &= \hat{w}_t - \hat{w}_{t-1} + \pi_t \\
r_t &= \phi_\pi \pi_t \\
\pi_t^w &= \frac{\kappa_w}{1+\epsilon_w \varphi}\left(\sigma c_t + \varphi n_t - \hat{w}_t\right) + \beta \mathbb{E}_t \pi_{t+1}^w - \gamma_w(\beta \pi_t - \pi_{t-1}) \\
y_t &= a_t + n_t \\
\hat{w}_t &= a_t \\
y_t &= c_t
\end{aligned}
$$